



WEB DEV INTERNSHIP

WEB DEVELOPMENT INTERNSHIP

Development of NOVA: A Responsive Registration Form with Client-Side Validation

STUDENT

Vibhav Mishra • Roll No. 25SCS1003001169

PROGRAMME

B.Tech CSE • Section 2CSE10

ORGANIZATION

Codec Technologies Pvt. Ltd. • Web Developer Intern

INTERNSHIP PERIOD

14 July 2026 – 14 August 2026

The screenshot shows a web application interface for a registration form. The header includes the NOVA logo, version (v1.0.0), and navigation links for Registration, Telemetry, and Documentation. The main content area is titled 'WEB DEVELOPMENT INTERNSHIP - CODEC TECHNOLOGIES' and features a large heading 'Build your next chapter.' with a subheading 'A high-performance developer onboarding portal engineered with clean semantic HTML5, reactive CSS3 glassmorphism, and real-time client-side validation logic.' Below this, there is a 'NVA ACCESS ID' section with a 'LIVE DEMO' button. The form itself is titled 'ACCOUNT ONBOARDING' and 'Create your account'. It includes a 'FORM COMPLETION' progress bar and several input fields: Full Name (with a placeholder 'e.g. Vibhav Mishra'), Email Address (with a placeholder 'vibhav@example.com'), Phone Number (with a placeholder '10-digit mobile number'), Date of Birth (with a placeholder 'dd-mm-yyyy'), Gender (with a 'Select gender' dropdown), Course / Program (with a 'Choose a course' dropdown), and City (with a placeholder 'e.g. New Delhi'). There are also fields for Password and Confirm Password, each with a 'Show' button. At the bottom, there is a 'Complete Registration' button and a link to 'I agree to the terms and conditions and data privacy guidelines.'

Actual NOVA Interface

Internship Overview

This one-month internship provided hands-on, industry-oriented exposure to frontend web development, allowing classroom concepts of HTML, CSS and JavaScript to be applied to a real, complete web interface — NOVA.



Practical Exposure

Moved from academic theory to a fully working, real-world frontend project.



Applied Learning

Directly applied HTML5, CSS3 and JavaScript concepts learned in the CSE programme.



Structured Progress

Followed a four-week, milestone-based project development approach.

INTERNSHIP DETAILS

Organization	Codec Technologies Pvt. Ltd.
Role	Web Developer Intern
Duration	14 July 2026 – 14 August 2026
Mode	Hybrid
Location	Pan India
Domain	Web Development / Frontend
Reporting To	Assigned Project Head(s)

About Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. is the organization at which this internship was completed. As per the official offer letter and internship certificate, Codec Technologies is a platform dedicated to empowering learners and connecting talent with meaningful career opportunities, offering IT and business consultancy services. The program is affiliated with the National Internship Portal and is recognized as an AICTE & ICAC approved internship.

DESIGNATION

Web Developer Intern

MODE

Hybrid

LOCATION

Pan India

REPORTING STRUCTURE

Assigned Project Head(s)

PROGRAM RECOGNITION

- ★ 1-Month AICTE & ICAC Approved Internship Program
- ★ Listed on the National Internship Portal (Ministry of Education)
- ★ Google for Education Partner organization
- ★ Performance reviewed on a weekly basis by project heads

Internship Objectives & Scope

OBJECTIVES

- ▶ Gain practical, hands-on frontend development exposure
- ▶ Apply HTML5, CSS3 and JavaScript in a real project
- ▶ Understand end-to-end web interface development
- ▶ Implement responsive design and client-side validation
- ▶ Apply core UI/UX principles
- ▶ Practice debugging, testing and documentation

SCOPE

- ▶ Frontend interface development (HTML5 / CSS3 / JS)
- ▶ Registration form design and structuring
- ▶ Responsive layout across screen sizes
- ▶ JavaScript-based interactivity
- ▶ Client-side validation logic
- ▶ Manual testing and UI refinement

Weekly Internship Progress

1

WEEK 1

Foundation & Setup

- Orientation & requirement understanding
- Development environment setup
- HTML5 & CSS3 fundamentals
- Project planning

2

WEEK 2

UI Development

- Registration form structure
- CSS styling & layout
- Typography & responsive basics
- NOVA interface build-out

3

WEEK 3

JavaScript & Validation

- DOM interaction & event handling
- Client-side validation rules
- Inline error messages
- Password visibility toggle

4

WEEK 4

Testing & Finalization

- Validation & responsive testing
- Debugging & UI refinement
- Documentation
- Presentation preparation

Project Introduction — NOVA

NOVA — Responsive Registration Form

NOVA is a modern, responsive frontend registration interface developed using HTML5, CSS3 and Vanilla JavaScript. It delivers a structured, visually clear and interactive registration experience with real-time client-side validation and immediate user feedback.

MAIN FOCUS AREAS

Responsive UI

Structured Form

Interactive Controls

Client-Side Validation

Clear Error Feedback

User-Friendly Flow

The screenshot displays the NOVA registration interface. The header includes the NOVA logo, version 'v2.0 SaaS', and navigation links for 'Registration', 'Telemetry', and 'Documentation'. A status indicator shows 'Codec Technologies - Verified Intern'. The main content area is titled 'Build your next chapter.' and describes the portal as a high-performance developer onboarding tool. It features a 'WEB DEVELOPMENT INTERNSHIP · CODEC TECHNOLOGIES' header and a 'LIVE SYNC' button. The form is divided into two main sections: 'ACCOUNT ONBOARDING' and 'Create your account'. The 'ACCOUNT ONBOARDING' section includes a 'FORM COMPLETION' progress bar at 8%, a 'Full Name' field with the value 'e.g. Vibhav Mishra', an 'Email Address' field with 'vibhav@example.com', a 'Phone Number' field with '10-digit mobile number', a 'Date of Birth' field with 'dd-mm-yyyy', a 'Gender' dropdown with 'Select gender', a 'Course / Program' dropdown with 'Choose a course', and a 'City' dropdown with 'e.g. New Delhi'. The 'Create your account' section includes a 'Password' field with 'Min. 8 characters' and a 'Show' button, and a 'Confirm Password' field with 'Repeat password' and a 'Show' button. The bottom of the form features three statistics: '10 VALIDATED FIELDS', '100% RESPONSIVE UI', and 'JS CLIENT REGEX ENGINE'.

Problem Statement & Project Objectives



THE PROBLEM

- ✗ Poor visual organization in basic registration forms
- ✗ Weak or missing input validation
- ✗ Unclear feedback when errors occur
- ✗ Poor responsiveness across screen sizes
- ✗ Low interactivity and unclear user guidance



PROJECT OBJECTIVES

- ✓ Build a modern, well-organized registration UI
- ✓ Create a fully responsive layout
- ✓ Implement robust client-side validation
- ✓ Provide immediate, field-level feedback
- ✓ Apply HTML5, CSS3 & JavaScript knowledge practically

Tools & Technologies Used



HTML5

- Page & form structure
- Semantic organization
- All registration input fields



CSS3

- Layout & spacing
- Typography & visual styling
- Responsive design behaviour



Vanilla JavaScript

- Client-side form validation
- DOM interaction & events
- Password visibility, progress bar, success/reset behaviour

Key Features of NOVA

REGISTRATION FIELDS

Full Name

Email Address

Phone Number

Date of Birth

Gender

Course / Program

City

Password

Confirm Password

Terms & Conditions

INTERACTIVE FEATURES

- Client-side validation
- Inline error messages
- Password show/hide
- Live progress indicator
- Success state
- Reset / register another
- Responsive UI

The screenshot displays the NOVA registration interface. On the left, a sidebar titled 'Build your next chapter.' provides context about the portal. The main area is titled 'ACCOUNT ONBOARDING' and 'Create your account'. A progress bar at the top right indicates 'FORM COMPLETION' at 100%. The form fields are filled with the following data: Full Name (vibhav m.), Email Address (vibhavhost@gmail.com), Phone Number (825423446), Date of Birth (01-01-2005), Gender (Male), Course / Program (B.Tech - Computer Science), and City (New Delhi). The Password and Confirm Password fields are masked with dots. A 'Completed Registration' button is visible at the bottom right. The interface includes various UI elements like a 'Real-time Validation' badge, a 'Live Sync' button, and a 'Zero-Lag Validation' indicator.

Completed Registration Form

UI/UX & Responsive Design

- ✓ Clear visual hierarchy
- ✓ Consistent spacing
- ✓ Readable typography
- ✓ Strong contrast
- ✓ Logical field grouping
- ✓ Simple, guided user flow

RESPONSIVE BEHAVIOUR

Desktop

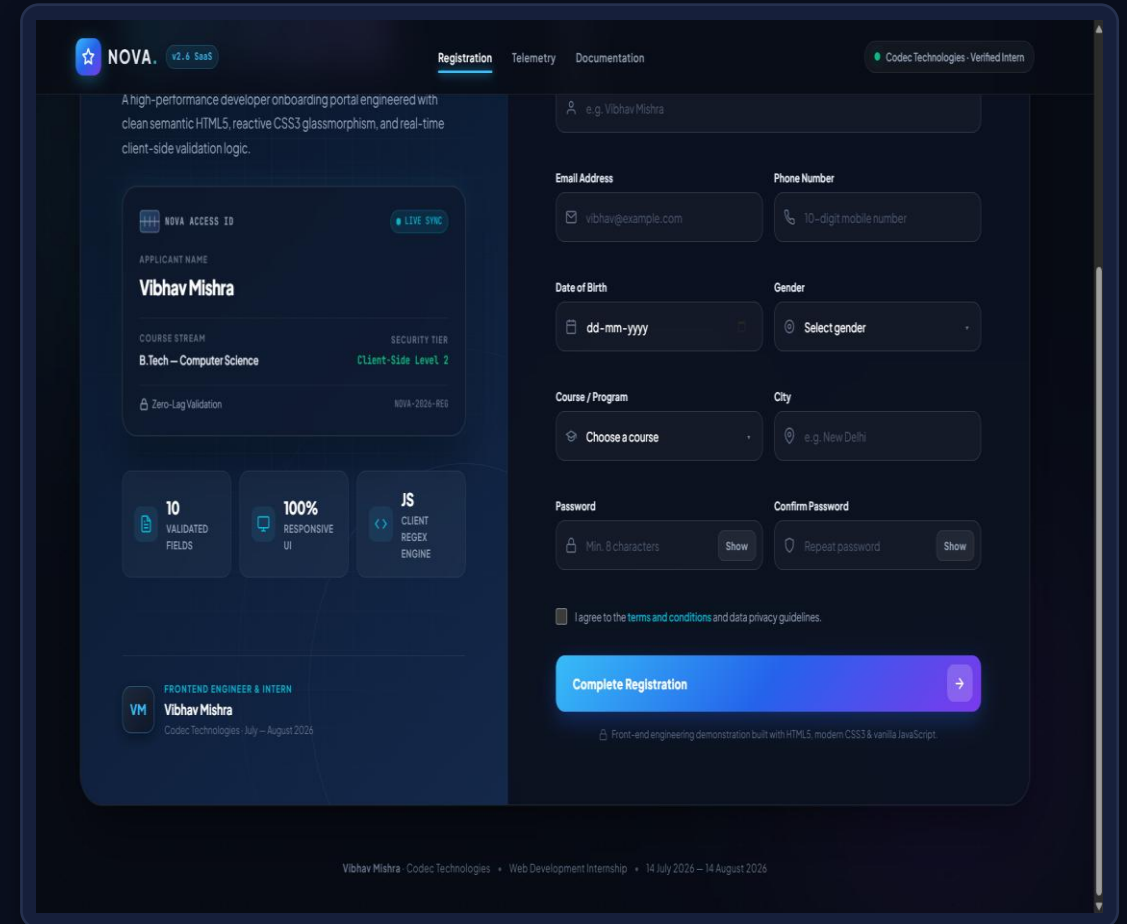


Tablet



Mobile

The layout adapts its structure and spacing to maintain clarity and usability across different screen sizes, while preserving the same validation logic and visual identity.



Full-Page Responsive Interface — Actual Screenshot

Form Validation & Project Workflow

VALIDATION LOGIC

JavaScript validates every field before the form can reach the successful registration state.

- Required full name
- Valid email format
- Valid 10-digit phone number
- Minimum password length
- Confirm password match
- Terms & Conditions accepted

Invalid → Error

Valid → Success

WORKFLOW

- 1 Open NOVA
- 2 Enter Personal Info
- 3 Enter Account Details
- 4 Accept Terms
- 5 Complete Registration
- 6 JS Validation
- 7 Error / Success

Client-Side Validation Error State

Implementation & Project Structure

PROJECT FILES

NOVA Registration Form

Project root

- `index.html` *Structure & form elements*
- `style.css` *Styling, layout & responsive design*
- `script.js` *Validation & interactive behaviour*
- `README.md` *Project documentation*

FRONTEND ARCHITECTURE



Testing, Results & Project Demonstration

TESTING

- Empty required fields
- Invalid email format
- Invalid phone number
- Weak / mismatched password
- Terms not accepted
- Valid, complete form

RESULT

NOVA reliably delivers a responsive interface with real-time validation, inline feedback, interactive controls, a success state and reset functionality.

The screenshot shows the 'Create your account' form with the following fields: Full Name, Email Address, Phone Number, Date of Birth, Gender, Course / Program, City, Password, and Confirm Password. The progress bar at the top indicates 0% completion.

Main Interface

The screenshot shows the same registration form, but with red error messages indicating 'Invalid email format' and 'Invalid phone number'. The progress bar remains at 0%.

Validation Error

The screenshot shows the registration form with all fields filled out. The progress bar at the top indicates 100% completion. The 'Complete Registration' button is highlighted.

Completed Form

The screenshot shows the 'Registration successful' message. Below it, a table displays the validated registration record:

VALIDATED REGISTRATION RECORD			
Full Name	EMAIL	PHONE	COURSE
vibhav m.	vibhavm@gmail.com	8254123446	B.Tech - Computer Science

The 'Register another account' button is visible at the bottom.

Success State

Learning Outcomes, Conclusion & Future Scope

TECHNICAL SKILLS

HTML5 • CSS3 • Responsive Design • JavaScript • DOM
Interaction • Form Validation • UI/UX • Testing &
Debugging • Documentation

PROFESSIONAL SKILLS

Time Management • Requirement Understanding •
Problem Solving • Project Planning • Presentation Skills

CONCLUSION

This one-month Web Development internship enabled practical application of HTML5, CSS3 and JavaScript through NOVA, resulting in a responsive, interactive registration experience and reinforcing the bridge between academic learning and real-world development.

FUTURE SCOPE

Backend integration • Database connectivity • Authentication •
Server-side validation • Email verification • User dashboard •
Full deployment

THANK YOU

Vibhav Mishra

Web Developer Intern

Codec Technologies Pvt. Ltd.